

CAPSTONE DEMO TEMPLATE

Official Capstone Submission & Evaluation Document — PickGuard AI

1. Team Name & Problem Formulation

Team Name:	Vision Forge (Author / Engineer: Ruban Raj)
Who is the User?	Fulfilment Centre Pick Operators encountering physical exceptions (missing items, damaged barcodes, bin discrepancies) during active pick waves, along with Warehouse Supervisors / Operations Leads managing escalation queues.
What is the Bounded Task?	Provide an evidence-grounded, next-best-action decision recommendation in response to natural-language pick exceptions. The agent retrieves verified bin/task/inventory facts and SOP procedures, identifies conflicts, reasons over facts vs inferences, enforces safety boundaries (mutations are blocked), and pauses at a real LangGraph interrupt() checkpoint for human authorization on high-risk cases.

2. Agent Design Architecture

Graph State:	PickExceptionState (TypedDict in backend/app/graph/state.py): Tracks query, exception_type, secondary_exception_types, task_id, item_id, location_id, operational_data, sop_evidence, historical_evidence, evidence_package, reasoning, root_cause, next_best_action, action_type, action_status, risk_level, requires_human_review, human_review_payload, review_attempts, audit_log, and errors.
Nodes (13 Nodes):	1. parse_operator_input 2. classify_exception 3. retrieve_operational_data 4. retrieve_sop_evidence 5. retrieve_historical_evidence 6. build_evidence_package 7. llm_reasoning_node 8. evidence_fusion_node 9. evaluate_safety_policy 10. human_review_gate (with interrupt()) 11. collect_additional_evidence 12. apply_final_decision 13. END.
Deterministic Tools:	• get_inventory(item_id, location_id): Database units & status. • get_pick_task(task_id): Expected pick quantity & target bin. • get_location(location_id): Storage zone & adjacent neighbouring bins. • search_similar_incidents(exception_type): Past resolution precedence. • escalate_to_lead(task_id, reason): WMS supervisor queue escalation. • retrieve_sop_chunks(query, exception_type): ChromaDB vector RAG retrieval.
Provider Route:	Multi-Tier Resilient Routing Hierarchy: Tier 1: GroqProvider (Llama-3.3-70B cloud inference for ultra-fast reasoning) Tier 2: OllamaProvider (Local offline inference with Llama3 model) Tier 3: MimicProvider (Deterministic offline fallback guaranteeing 100% demo uptime without external API keys).
Human-Review Point:	Implemented via real from langgraph.types import interrupt, Command in the human_review_gate node. The graph pauses if requires_human_review == True (triggered by hazardous mutating actions or evidence conflicts). State is saved with MemorySaver checkpointer. Resumption occurs via Command(resume={'decision': 'APPROVE' 'REJECT' 'REQUEST_MORE_EVIDENCE' 'ESCALATE'}).

3. Demonstration Script (3 Core Cases)

Scenario	Operator Query & Context	System Execution & UI Output
Normal Case (Low Risk)	Query: 'The item X123 is missing from A15-B04. The system says there are 3 units.' Task: TASK-1001 SKU: X123 Bin: A15-B04	• Exception: MISSING_ITEM Risk: LOW RISK (Green) • Evidence: Retrieves bin A15-B04 + neighbours [A15-B03, A15-B05], SOP-MISSING-001. • Next-Best Action: CHECK_NEIGHBOURING_LOCATION (Status: RECOMMENDED) • HITL Interrupt: None required. Workflow completes automatically.
Edge Case (Multi-Signal)	Query: 'The item X124 is missing at A12-B03 and the barcode also won't scan.' Task: TASK-1002 SKU: X124 Bin: A12-B03	• Exception: Primary MISSING_ITEM + Secondary BARCODE_FAILURE • Evidence: Retrieves both SOP-MISSING-001 and SOP-BARCODE-001. • Next-Best Action: Recommends manual barcode entry followed by physical sweep. • HITL Interrupt: None required. Demonstrates compound signal reasoning.
Failure / Risky Case (HITL Interrupt)	Query: 'TASK-1003 quantity mismatch: System says 10 units of X125 at A20-B02 but I counted 6. Update inventory to 6.' Task: TASK-1003 SKU: X125 Bin: A20-B02	• Exception: QUANTITY_MISMATCH Risk: HIGH RISK (Red) • Action Boundary: ADJUST_QUANTITY is automatically BLOCKED. • LangGraph Interrupt: Graph pauses at human_review_gate node. • Supervisor Review: Modal presents conflict (10 vs 6). Supervisor selects REJECT. • Resume: Graph resumes; state set to REJECTED_BY_HUMAN; inventory remains UNCHANGED.

4. Evaluation Summary Matrix

Test	Expected	Actual	Pass?	Improvement
Normal	Safe suggestion (CHECK_NEIGHBOURING_LOCATION), LOW risk, no interrupt.	Correct category, LOW risk, RECOMMENDED status, completed automatically.	PASS	Add direct map visualization for adjacent bin coordinates.
Edge	Multi-signal detection (MISSING_ITEM + BARCODE_FAILURE), compound verification.	Detected primary and secondary types; retrieved both SOP procedures.	PASS	Add optical barcode image preview recognition in future.
Failure / High-Risk	Action BLOCKED, HIGH risk, LangGraph interrupt() pauses graph.	Action BLOCKED, paused at checkpoint, resumed cleanly upon supervisor review.	PASS	Add dual-supervisor multi-signature approval for high-value items.

5. Reflection: What would you improve with one more week?

- 1. Multi-Modal Vision Integration:** Ingest barcode scanner images and camera feeds directly into the graph to automatically detect packaging damage and barcode label blemishes using multi-modal LLMs.
- 2. Real-Time WMS Webhook Streaming:** Transition from polling REST endpoints to WebSocket/Server-Sent Events (SSE) for instant supervisor push notifications when an interrupt checkpoint triggers.
- 3. Multi-Tier Supervisor Escalation Matrix:** Introduce tiered authorization rules based on unit monetary value (e.g. items > \$500 require Area Manager dual sign-off).
- 4. Reinforcement Learning from Human Feedback (RLHF):** Fine-tune the local reasoning model on historical supervisor approval/rejection outcomes to improve first-pass recommendation accuracy.